

Assignment No:5

Topic : Inheritance, Abstract Class

1. Write a program that creates a class Account that have members customer name, account number. Account has constructor to initialize its members and method display() to show the result. Create a child class Savings_Account that is derived from Account class. Savings_Account have members min_bal and saving_bal. Use show() to display its details. Then create another child class Account_details from Savings_Account class have members deposit, withdrawl and a method show1() to show its details. Child classes have constructors of their own. Create a driver class that creates a record of customer and display all its details.

SOLUTION:

```
import java.util.Scanner;
class Account {
    String customerName;
    int accountNumber;
    public Account(String customerName, int accountNumber) {
        this.customerName = customerName;
        this.accountNumber = accountNumber;
    }
    public void display() {
        System.out.println("Customer Name: " + customerName);
        System.out.println("Account Number: " + accountNumber);
    }
}
class Savings_Account extends Account {
    double min_bal;
```

```
double saving_bal;

public Savings_Account(String customerName, int accountNumber, double min_bal,
double saving_bal) {

    super(customerName, accountNumber);

    this.min_bal = min_bal;

    this.saving_bal = saving_bal;

}

public void show() {

    display();

    System.out.println("Minimum Balance: " + min_bal);

    System.out.println("Savings Balance: " + saving_bal);

}

}

class Account_details extends Savings_Account {

    double deposit;

    double withdrawal;

    public Account_details(String customerName, int accountNumber, double min_bal,
double saving_bal, double deposit, double withdrawal) {

        super(customerName, accountNumber, min_bal, saving_bal);

        this.deposit = deposit;

        this.withdrawal = withdrawal;

    }

    public void show1() {

        show();

    }

}
```

```
        System.out.println("Deposit Amount: " + deposit);

        System.out.println("Withdrawal Amount: " + withdrawal);
    }
}

public class AccountDriver {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter customer name: ");

        String customerName = sc.nextLine();

        System.out.print("Enter account number: ");

        int accountNumber = sc.nextInt();

        System.out.print("Enter minimum balance: ");

        double min_bal = sc.nextDouble();

        System.out.print("Enter savings balance: ");

        double saving_bal = sc.nextDouble();

        System.out.print("Enter deposit amount: ");

        double deposit = sc.nextDouble();

        System.out.print("Enter withdrawal amount: ");

        double withdrawal = sc.nextDouble();

        Account_details customer = new Account_details(customerName, accountNumber,
min_bal, saving_bal, deposit, withdrawal);

        System.out.println("\nAccount Details of the Customer:");

        customer.show1();

        sc.close();
    }
}
```

```
}  
}
```

2. Create a class Figure with instance members dim1 and dim2. Use constructor and area() that returns the area of figure. Create a derived class Rectangle derived from Figure and area() that returns the area of rectangle. Create another derived class Triangle and Square that has area() which returns area of Triangle and Square respectively. Derived class have appropriate constructor. Using method overriding concept test the functionalities of derived class by creating objects and super class memory references in Driver class.

SOLUTION:

```
import java.util.Scanner;  
class Figure {  
    double dim1;  
    double dim2;  
    public Figure(double d1, double d2) {  
        dim1 = d1;  
        dim2 = d2;  
    }  
    double area() {
```

```
System.out.println("Area of Figure is undefined."); return 0;
```

```
}
```

```
}
```

```
class Rectangle extends Figure {
```

```
    public Rectangle(double length, double width) {
```

```
        super(length, width);
```

```
    }
```

```
    double area() {
```

```
        return dim1 * dim2;
```

```
    }
```

```
}
```

```
class Triangle extends Figure {
```

```
    public Triangle(double base, double height) {
```

```
        super(base, height);
```

```
    }
```

```
    double area() {
```

```
        return 0.5 * dim1 * dim2;
```

```
    }
```

```
}
```

```
class Square extends Figure {
```

```
    public Square(double side) {
```

```
        super(side, side);
```

```

        super(side, side);
    }

    double area() {
        return dim1 * dim2;
    }
}

public class Driver {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Figure fig = null;

        System.out.println("Choose Shape: ");

        System.out.println("1. Rectangle");

        System.out.println("2. Triangle");

        System.out.println("3. Square");

        int choice = sc.nextInt();

        switch (choice) {

            case 1:

                System.out.print("Enter length: ");

                double length = sc.nextDouble();

                System.out.print("Enter width: ");

                double width = sc.nextDouble();

                fig = new Rectangle(length, width);

                break;

```

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case 2:

```
System.out.print("Enter base: ");  
  
double base = sc.nextDouble();  
  
System.out.print("Enter height: ");  
  
double height = sc.nextDouble();  
  
fig = new Triangle(base, height);  
  
break;
```

case 3:

```
System.out.print("Enter side: ");  
  
double side = sc.nextDouble();  
  
fig = new Square(side);  
  
break;
```

default:

```
System.out.println("Invalid choice");  
  
sc.close();  
  
return;
```

```
}
```

```
System.out.println("Area: " + fig.area());
```

```
sc.close();
```

```
}
```

```
}
```

3. Write a program to create a class named Shape. It should contain two methods, draw() and erase() that prints "Drawing Shape" and "Erasing Shape" respectively. For this class, create three sub classes, Circle, Triangle and Square and each class should override the parent class functions - draw () and erase (). The draw() method should print "Drawing Circle", "Drawing Triangle" and "Drawing Square" respectively. The erase() method should print "Erasing Circle", "Erasing Triangle" and "Erasing Square" respectively. Create objects of Circle, Triangle and Square, assign each to Shape variable(reference) and call draw() and erase() method using each object.

SOLUTION:

```
import java.util.Scanner;

class Shape {

    void draw() {

        System.out.println("Drawing Shape");

    }

    void erase() {

        System.out.println("Erasing Shape");

    }

}

class Circle extends Shape {

    @Override

    void draw() {
```



```
        System.out.println("Drawing Circle");
    }

    @Override
    void erase() {
        System.out.println("Erasing Circle");
    }
}

class Triangle extends Shape {

    @Override
    void draw() {
        System.out.println("Drawing Triangle");
    }

    @Override
    void erase() {
        System.out.println("Erasing Triangle");
    }
}

class Square extends Shape {

    @Override
    void draw() {
        System.out.println("Drawing Square");
    }
}
```

```
@Override  
  
void erase() {  
  
    System.out.println("Erasing Square");  
  
}  
  
}  
  
public class Drawing {  
  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        Shape shape = null;  
  
  
        System.out.println("Choose Shape to Draw and Erase:");  
  
        System.out.println("1. Circle");  
  
        System.out.println("2. Triangle");  
  
        System.out.println("3. Square");  
  
        int choice = sc.nextInt();  
  
        switch (choice) {  
  
            case 1:  
  
                shape = new Circle();  
  
                break;  
  
            case 2:  
  
                shape = new Triangle();  
  
                break;  
  
            case 3:
```

```

        shape = new Square();

        break;

    default:

        System.out.println("Invalid choice");

        sc.close();

        return;

    }

    shape.draw();

    shape.erase();

    sc.close();

}

}

```

4. Define an abstract class named “Figure”, having data members dim1 and dim2. Extend this class to create two concrete classes named Rectangle and Triangle. Override the getArea() method in the sub classes. Invoke the getArea() method in the main method of another Driver class through the abstract class reference variable.

SOLUTION:

```
import java.util.Scanner;
```

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```
abstract class Figure {  
    double dim1;  
    double dim2;  
    public Figure(double d1, double d2) {  
        dim1 = d1;  
        dim2 = d2;  
    }  
    abstract double getArea();  
}  
  
class Rectangle extends Figure {  
    public Rectangle(double length, double width) {  
        super(length, width);  
    }  
  
    @Override  
    double getArea() {  
        return dim1 * dim2;  
    }  
}  
  
class Triangle extends Figure {  
    public Triangle(double base, double height) {  
        super(base, height);  
    }  
}
```

```

@Override

double getArea() {

    return 0.5 * dim1 * dim2;

}

}

public class Areafig {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Figure fig = null;

        System.out.println("Choose Shape to Calculate Area:");

        System.out.println("1. Rectangle");

        System.out.println("2. Triangle");

        int choice = sc.nextInt();

        switch (choice) {

            case 1:

                System.out.print("Enter length: ");

                double length = sc.nextDouble();

                System.out.print("Enter width: ");

                double width = sc.nextDouble();

                fig = new Rectangle(length, width);

                break;

            case 2:

                System.out.print("Enter base: ");

                double base = sc.nextDouble();

```

```

        System.out.print("Enter height: ");

        double height = sc.nextDouble();

        fig = new Triangle(base, height);

        break;

    default:

        System.out.println("Invalid choice");

        sc.close();

        return;

    }

    System.out.println("Area: " + fig.getArea());

    sc.close();

}

}

```

5. Create a class Point2D with the data member x and y coordinate. Use default and parameterised constructor to set the coordinate values and display() to show the coordinates. Create a subclass called Point3D which is derived from the superclass Point2D with data members z coordinate and has constructor to initialize the input and show() method to display the coordinates. Test the methods of both the classes by creating objects in the main method of driver class.

SOLUTION:

```

import java.util.Scanner;

class Point2D {

    protected double x;

```

```
        protected double y;

    public Point2D() {

        x = 0;

        y = 0;

    }

    public Point2D(double x, double y) {

        this.x = x;

        this.y = y;

    }

    public void display() {

        System.out.println("Point2D Coordinates: (" + x + ", " + y + ")");

    }

}

class Point3D extends Point2D {

    private double z;

    public Point3D(double x, double y, double z) {

        super(x, y);

        this.z = z;

    }

    // Method to show 3D coordinates

    public void show() {

        System.out.println("Point3D Coordinates: (" + x + ", " + y + ", " + z + ")");

    }

}
```

```
}

public class PointDriver {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter x coordinate for Point2D: ");

        double x2d = sc.nextDouble();

        System.out.print("Enter y coordinate for Point2D: ");

        double y2d = sc.nextDouble();

        Point2D p2d = new Point2D(x2d, y2d);

        p2d.display();

        // Input for Point3D

        System.out.print("Enter x coordinate for Point3D: ");

        double x3d = sc.nextDouble();

        System.out.print("Enter y coordinate for Point3D: ");

        double y3d = sc.nextDouble();

        System.out.print("Enter z coordinate for Point3D: ");

        double z3d = sc.nextDouble();

        Point3D p3d = new Point3D(x3d, y3d, z3d);

        p3d.show();

        sc.close();

    }

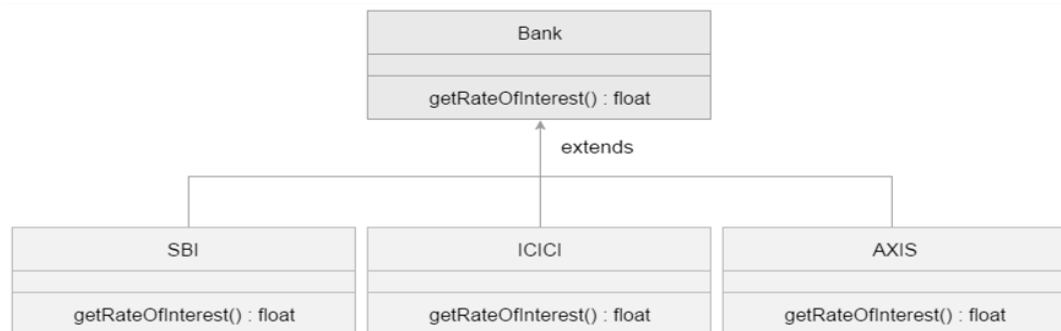
}
```


6. Create the classes as given in the below figure. Display the interest rate in the following format:

SBI Rate of Interest : 8

ICICI Rate of Interest : 7

AXIS Rate of Interest : 9



SOLUTION:

```
import java.util.Scanner;
class Bank {
    public float getRateOfInterest() {
        return 0;
    }
}
class SBI extends Bank {
    public float getRateOfInterest() {
        return 8;
    }
}
class ICICI extends Bank {
    public float getRateOfInterest() {
        return 7;
    }
}
```

```
}  
class AXIS extends Bank {  
    public float getRateOfInterest() {  
        return 9;  
    }  
}  
  
public class BankInterest {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.println("Enter Bank Name (SBI / ICICI / AXIS): ");  
        String bankName = sc.nextLine().trim().toUpperCase();  
  
        Bank b;  
        switch (bankName) {  
            case "SBI":  
                b = new SBI();  
                break;  
            case "ICICI":  
                b = new ICICI();  
                break;  
            case "AXIS":  
                b = new AXIS();  
                break;  
            default:  
                System.out.println("Invalid bank name entered!");  
                sc.close();  
                return;  
        }  
  
        System.out.println(bankName + " Rate of Interest: " + b.getRateOfInterest());  
        sc.close();  
    }  
}
```

```
}  
}
```

7. Create a class Person that has data member name. Use constructor to initialize name and display() to display name. Create a derived class Employee from Person class having private members empid. Using constructor initialize empid and have method display() to display empid. Create another derivedclass HourlyEmployee from Employee with private members hourlyRate and hoursWorked. Use constructor to initialize input and methods getGrossPay() that computes and returns the gross pay of the employee and display() to display the hourlyRate, hoursWorked and gross pay. Create a driver class to test the functionalities of the above classes and display output in the following format

Name : John Smith

EmpID : 7569

Hourly Rate : 100

Hours worked : 2000

Gross pay : 200000

SOLUTION:

```
import java.util.Scanner;
```

```
class Person {  
    protected String name;
```

```
    public Person(String name) {  
        this.name = name;  
    }
```

```
    public void display() {  
        System.out.println("Name: " + name);  
    }
```

```
}
```

```
class Employee extends Person {
```

```
protected int empid;

public Employee(String name, int empid) {
    super(name);
    this.empid = empid;
}

@Override
public void display() {
    super.display();
    System.out.println("EmpID: " + empid);
}
}

class HourlyEmployee extends Employee {
    private double hourlyRate;
    private int hoursWorked;

    public HourlyEmployee(String name, int empid, double hourlyRate, int
hoursWorked) {
        super(name, empid);
        this.hourlyRate = hourlyRate;
        this.hoursWorked = hoursWorked;
    }

    public double getGrossPay() {
        return hourlyRate * hoursWorked;
    }

    @Override
    public void display() {
        super.display();
        System.out.println("Hourly Rate: " + hourlyRate);
    }
}
```

```
        System.out.println("Hours Worked: " + hoursWorked);
        System.out.println("Gross Pay: " + getGrossPay());
    }
}

public class EmployeeTestclass {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Employee ID: ");
        int empid = sc.nextInt();
        System.out.print("Enter Hourly Rate: ");
        double rate = sc.nextDouble();
        System.out.print("Enter Hours Worked: ");
        int hours = sc.nextInt();
        HourlyEmployee h = new HourlyEmployee(name, empid, rate, hours);
        System.out.println("\n--- Employee Details ---");
        h.display();
        sc.close();
    }
}
```